

OPENIMIS

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open**IMIS**

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Introduction



- An open-source software tool for the digitalisation and efficient management of health financing workflows
- Currently being used to manage a variety of health protection schemes in many countries including Nepal

Key Features



Social Protection to All

It has created an ecosystem which offers insurance social protection to all the people, who can't afford for other insurance policy.

Key Features



Reliable and Cost-effective Digital Solutions

- OpenMIS is an open source software, which can be built and set up for very minimum cost.
- No need to develop a costly software.

Key Features



Technical Backup

We can have technical support from anyone anywhere, about how to set up the software and how does it work.

Key Features



Flexibility

Due to the flexibility, the software can be adapted to the various needs of low- and middle-income countries on their path to global good.

Key Features



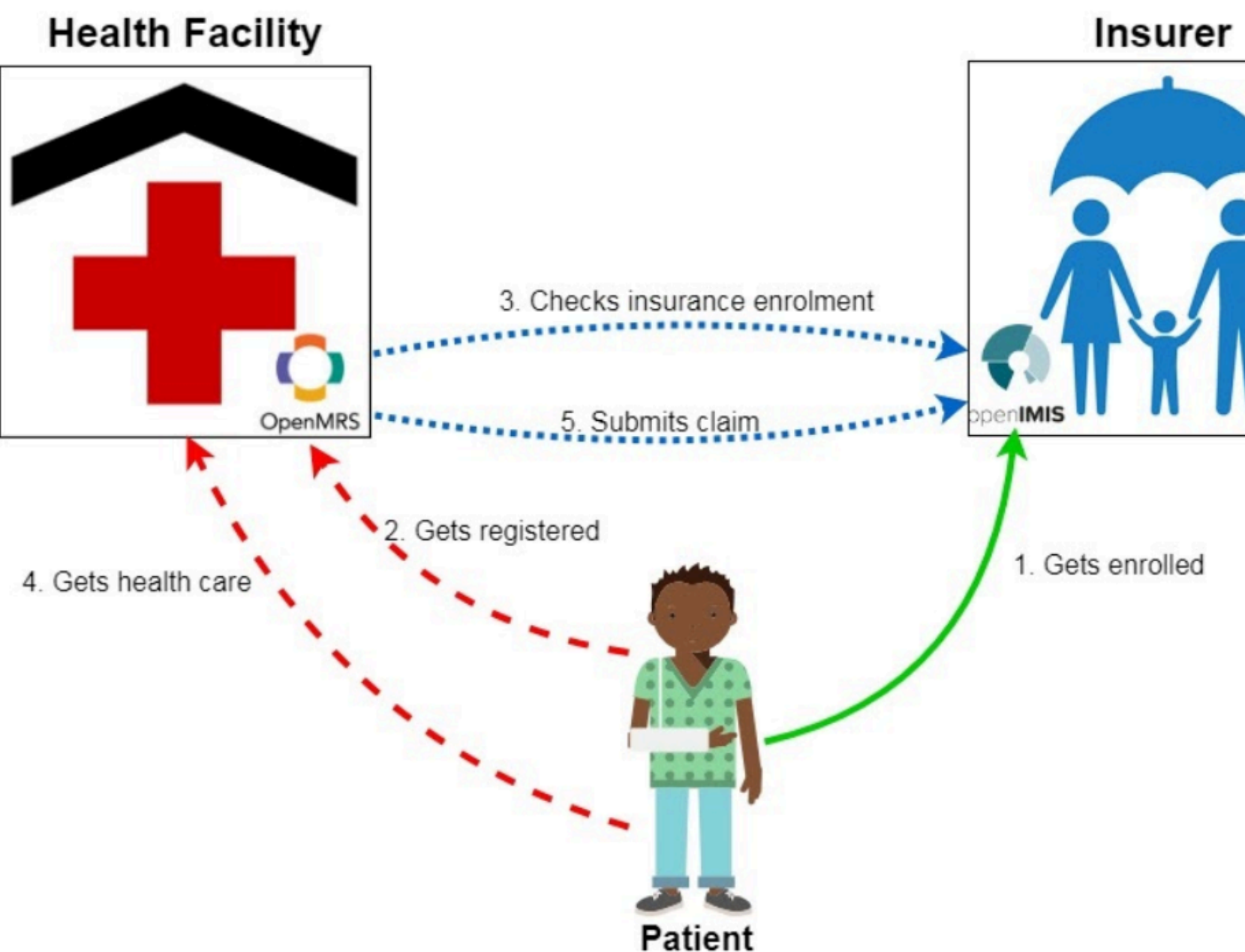
Modular Architecture

A structure for the continuous improvement through addition of new modules

Key Features

Easy Integration

openIMIS can be easily integrated with other opensource softwares like Bahmini, openMRS etc for better functioning.



Key Features



Opportunity

Offer opportunities for students, individuals and organisations to learn about openIMIS and become a service provider, e.g. as implementation partner or local support team



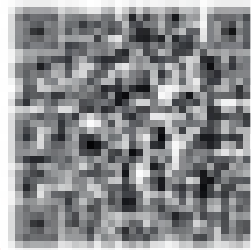
USE CASES IN NATIONAL LEVEL

USE CASE OF OPENIMIS

CHF membership card



District name:
Last name :
Other names :
Village :
Date of birth :
Sex: ☐ male ☐ female
Start date __ / __ / 20



ID no (Pre printed)

ALWAYS SHOW THIS CARD WHEN VISITING
A HEALTH FACILITY



1 ENROLLMENT



Household or individual



Paper or app registration



Data storage on server

- **Certain Individuals are given authority of Insurance agent by admin who adds the households or individuals as insured to the database through paper or app registration.**
- **In the context of Nepal, ward health volunteers (swyamsewak) are assigned to the roles of collecting and adding database.**



2 HEALTH SERVICE



Patient visit and verification



Patient treatment



Claim entry

- A person gets registered in a health facility and claims for insurance.
- OpenMIS based Client Registry is used by health facilities by contacting a server where all the data is stored.
- The claim is entered into the software, which is stored in the claimed model.



3 CLAIMS MANAGEMENT



Claim submissions



Claim review



Payment approval

- **OpenMIS searches for the claim in the insuree database table with the id and declaration date.**
- **OpenMIS sends information back to the server with info about fault or success of the process**
- **Depending on the fault or success of the process, the payment approval is done.**
- **In case of Nepal, payment is limited up to 1 lakh per family per year.**



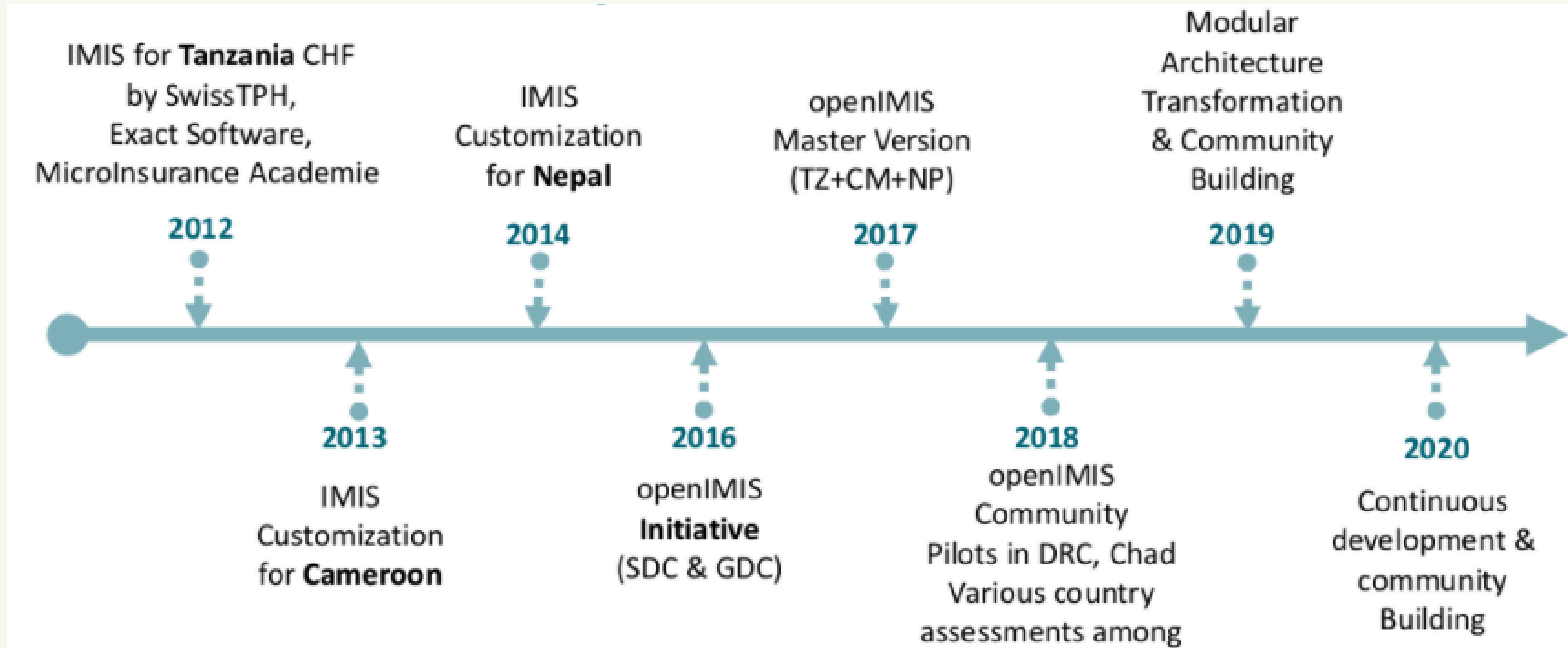
4 MONITORING AND REPORTING



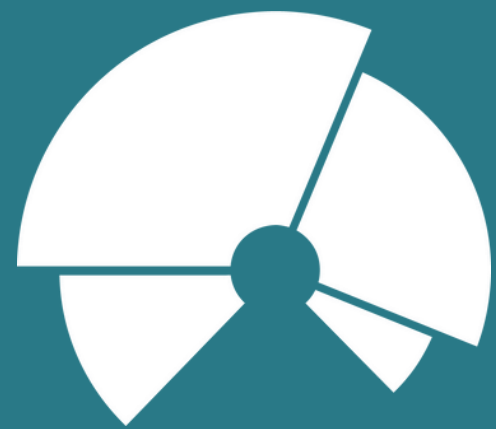
- **Monitoring and reporting of the data to the certain authority is done by the administrator.**
- **The data collected during the process can be used for the national health analysis.**
- **Analysis is further used in various health sector, it helps to keep record of health status of the individuals.**



Transition of openIMIS



OPENIMIS MODULAR ARCHITECTURE



SOLUTION LEVEL

Considering the solution as a coherent assembly of Software Components.

SOFTWARE COMPONENT LEVEL

Components are decomposed into modules dedicated to fulfil the expected scope for them

ENTITY LEVEL

Each concept hosted by a Software Component can be particularised to the specific needs of an implementation.



SOLUTION LEVEL

- OpenMIS is seen as a component of the OpenHIE platform.
- Modularity achieved by conforming to the OpenHIE platform standards
- By adopting the FHIR existing standards to expose OpenMIS - managed concepts





SOFTWARE COMPONENT AND ENTITY LEVELS

- Software Component and Entity Levels modularity are fully controlled by OpenMIS.
- From low-level features (like login) to central features (like Insurance Scheme management), all are built as plugins of the core OpenMIS platform.
- Deployments of OpenMIS doesn't impose a distributed system



Software Component Layers



Frontend



Backend



Database



Software Component Layers

FRONTEND

Modular and
Component Based
Application

BACKEND

Services Layer
Modular and
Transactional

DATABASE

Relational, with enforced
constraints (FKs, Unique
Key, Typed and
Mandatory Columns)



Transformation to Modular Architecture

ITERATIVE

A production-ready version of openMIS is delivered at each iteration

INCREMENTAL

Covering all original openMIS features (of each implementation) with plugins in the (new) openMIS



MODULAR ARCHITECTURE

- Both frontend and backend are based on two major principles: **contributions** and **events**
- **Contributions** prevent modules to require *deep knowledge* of others
- Modules don't call each others straight but **publish events** on which other modules *may* react



MODULAR ARCHITECTURE

- Programming with **contributions** and **events** more challenging and time-consuming than *traditional* programming
- Requires programmers to think more about modules' scope
- Requires a need to exchange more *generic* entities
- Requires the readiness to change code time and again
- **Contributions** and **Events** very important for customization

TECHNOLOGIES (FRONTEND)

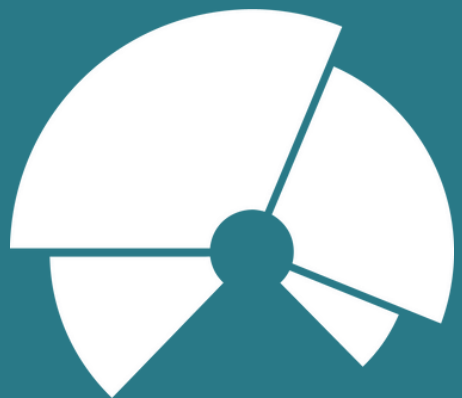
REACT COMPONENTS AND REDUX

Contributions mechanism is not provided *by default* in React but OpenMIS specific

Events implemented via Redux and the default dispatch/reducer mechanism

MODULES BUILT VIA ROLLUP

Applications assembled with webpack



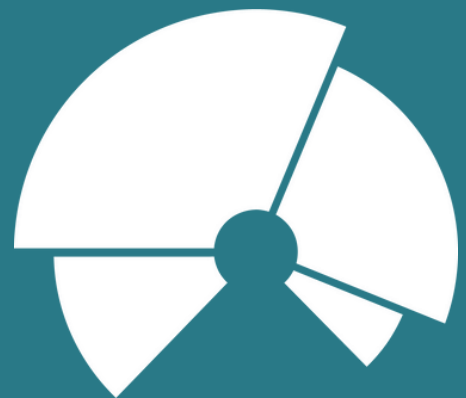


FRONTEND

The screenshot displays the 'openIMIS 1.2.1' application interface. The top navigation bar includes links for 'Insurees and Policies', 'Claims', 'Administration', 'Tools', and 'Profile', along with a search bar labeled 'Insuree enquiry...'. The main content area is titled '< Family/Group' and is divided into two sections: 'Family Details' and 'Head Insuree Details'. The 'Family Details' section contains fields for 'Region' (None), 'District' (None), 'Municipality' (with a search icon), 'Village' (with a search icon), 'Family Type' (None), 'Confirmation Type' (None), 'Confirmation No.', and 'Address details'. The 'Head Insuree Details' section contains fields for 'Insurance No. *', 'Last Name *', 'Given Names *', 'Birth Date *', 'Gender' (None), 'Marital Status' (Not Specified), a checkbox for 'Beneficiary Card', 'Photo Date', and 'Officer' (with a search icon).

- Each module contributes to the core module extension keys or another module extension key
- Modular format ensures no unnecessary dependency causing issues in the future
- Customized modules can be created using templates

FRONTEND MODULES



OPENIMIS-FE-CORE

- Holds the files of the openIMIS Frontend Core reference module.
- Provides openIMIS core mechanisms, openIMIS core components, openIMIS ModulesManager, many Generic components to be (re)used in other (business-focused) openimis components and various helpers.

OPENIMIS-FE-CLAIM

Holds the files of the openIMIS Frontend Claim reference module

FRONTEND MODULES

OPENIMIS-FE-MEDICAL

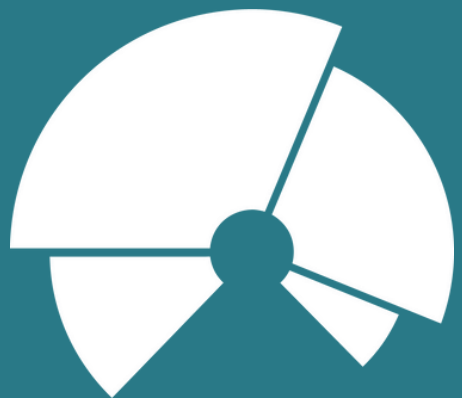
Contains frontend medical reference module

OPENIMIS-FE-PROFILE

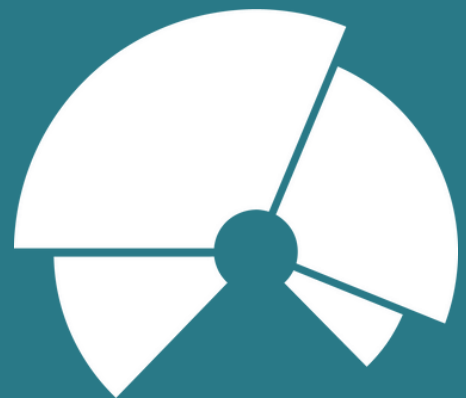
Contains frontend profile reference module

OPENIMIS-FE-TEMPLATE

Can customize your own module with this empty template



TECHNOLOGIES (BACKEND)



Django Models and Services

- **Events** in Django are called **Signals**

Default API in GraphQL (Graphene)

- **Queries** (synchronous)
- **Mutations** (asynchronous)
- **Contributions** in Django/Graphene enabled via a (openMIS) specific way of dynamically assembling the GraphQL schema

Virtual Environments (venv)

USERS (BACKEND)

Django administration

WELCOME, ADMIN. [VIEW SITE](#) / [CHANGE PASSWORD](#) / [LOG OUT](#)

Site administration

AUTHENTICATION AND AUTHORIZATION

Groups [+ Add](#) [Change](#)

Permissions [+ Add](#) [Change](#)

CORE

Module configurations [+ Add](#) [Change](#)

Technical users [+ Add](#) [Change](#)

ADMIN : Django Superuser

- The initial superuser must be created via the Django `createsuperuser`
- Admin can connect to application via `https://gateway/api/admin`.
- Admin can create new TechnicalUsers and groups to grant permissions to them



USERS (BACKEND)

REAL USERS

- Created by admin
- People who interact with openIMIS via a user interface to accomplish some insurance management tasks but has no direct access to backend.



SECURITY (BACKEND)

AUTHORIZATION LAYERS

- The Django default (basic) permissions mechanism
- The RBAC 'django-rules' extensions

If any of the 2 layers grants access, the access is given to the user.

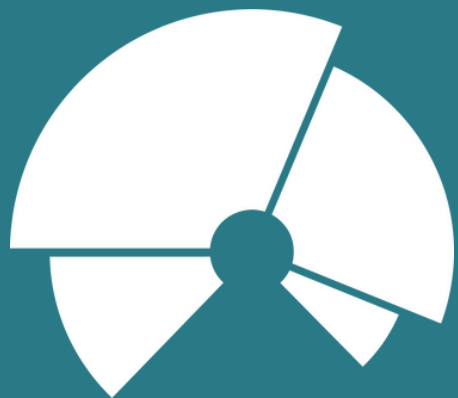


TECHNOLOGIES (DATABASE)

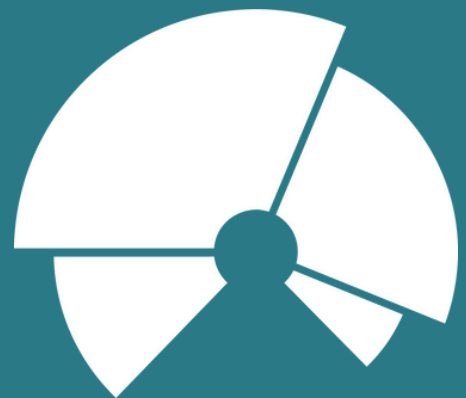
MS-SQL

- With constraints (fks, unique keys, mandatory columns)
- Managed via Django
 - Django Object Relation Mapping
 - Django Migrations
 - Using JSON fields for *extendable* objects

Being migrated to **PostgreSQL**



CONFIGURATION (DATABASE)



- Download and install the desired SQL server.
- Configure SQL Server
- Download the openMIS database SQL files and migration scripts
- Initialize openMIS database on your server.
- Create a dedicated user with full privilege on the openMIS database only.

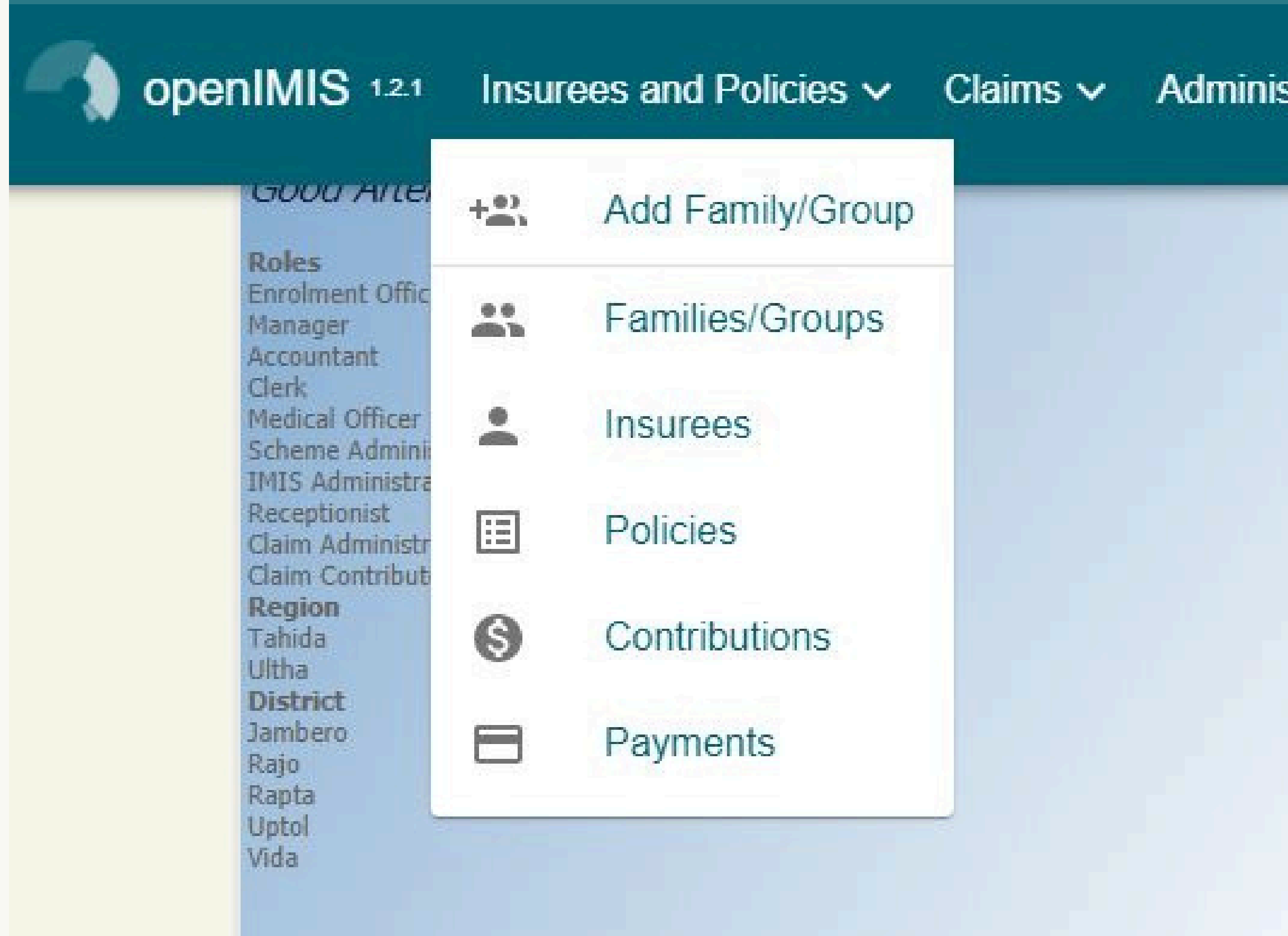


DEMO ADDING NEW FAMILIES AND POLICIES



The enrolment officer has the role of adding new families/groups. For this entry through the insurances and policies into add family/group option

ADDING A NEW FAMILY





The data entry window is to be filled with the appropriate details of the family head and a unique id (insurance no.) is given to each insuree. The family id is the same as the family head insurance id

ADDING A NEW FAMILY HEAD

openIMIS 1.2.1 Insurees and Policies ▾ Claims ▾ Administration ▾ Tools ▾ Profile ▾

Insuree enquiry...

< Family/Group Smith John (123456789)

Family Details

Region Q R1 Ultha X	District Q R1D1 Rapta X	Municipality Q R1D1M3 Adhi X	Village Q R1D1M3V1 Rathula X
Family Type Household ▾	Confirmation Type Municipality ▾	Confirmation No. 123e4r4321	Address details Rathulu ☐ Poverty Status

Head Insuree Details

Insurance No. * 123456789	Last Name * Smith	Given Names * John
Birth Date * 1980-08-12	Gender Male ▾	Marital Status Married ▾
	☐ Beneficiary Card	Photo Date * 2021-08-26
		Officer * Q E00004 Zombie X



The insuree detail window enlists all the details of the insuree. This window is utilized to add new policies and new members to a family.

INSUREE DETAILS

 **openIMIS** 1.2.1 Insurees and Policies ▾ Claims ▾ Administration ▾ Tools ▾ Profile ▾

[<](#) Family/Group Smith John (123456789)

Family Details

Region R1 Ultha	District R1D1 Rapta	Municipality R1D1M3 Adhi	Village R1D1M3V1 Rathulu
Head Ins. No. 123456789	Head Last Name Smith	Head Given Name John	Head Birth Date 1980-08-12
Family Type Household	Confirmation Type Municipality	Confirmation No. 123e4r4321	Address details Rathulu

1 Insurees

Insurance No.	Last Name	Given Names	Gender	Birth Date	Beneficiary Card
123456789	Smith	John	Male	1980-08-12	☐





Adding a member to a family can be achieved via the insuree details window of the family head. The members of the family are listed here.

ADDING A FAMILY MEMBER

1 Insurees

+ 

+

Insurance No.	Last Name	Given Names	Gender	Birth Date	Beneficiary Card
🔍 123456789	Smith	John	Male	1980-08-12	☐

Rows Per Page

5

1-1 of 1

<

>

1 Policies

☒ Show only not yet active, and last active or last suspended/expired

+

Product Code	Name	Expiry Date	Status	Deduction	Hospital Deduction	Non Hospital Deduction	Ceiling	Hospital Ceiling	Non Hospital Ceiling
BCUL0001	Basic Cover Ultha	2022-08-25	Idle				0	0	0

🔄





The family details of the member remain the same as the head, details as well as a unique insurance number is assigned to the new member

NEW MEMBER IN FAMILY

 **openIMIS** 1.2.1 Insurees and Policies ▾ Claims ▾ Administration ▾ Tools ▾ Profile ▾

< Insuree Smith Jenifer (123456788)

Family Details 

Region	District	Municipality	
R1 Ultha	R1D1 Rapta	R1D1M3 Adhi	
Head Ins. No.	Head Last Name	Head Given Name	
123456789	Smith	John	
Family Type	Confirmation Type	Confirmation No.	Address details
Household	Municipality	123e4r4321	Rathulu

Insuree

Insurance No. *	Last Name *
123456788	Smith



Policies for the entire family are added via the family head add policy option. For an individual policy of member, policy is added via insuree details window

ADD POLICIES

Policy details

Enrolment Date *	Effective Date	Start Date
2021-08-26		2021-11-01
Product *	Officer *	Status
FCUL0001 Fixed Cycle Cov X	E00001 Roberts X	Entered

Policy values

	Value	Contributions paid	Balance
	\$ 6000	\$ 2000	\$
		General	In-Patient
Deductible	\$	\$	
Remunerated Health Care	\$	\$	





open**IMIS**

CLAIM MODULE



CLAIM KEY FEATURES

- Verify patient data
- Add new claim
- Edit claim
- Print claim
- Delete Claim
- Submit Claim

Data Entry Page

- Mandatory fields are filled up.
- On submit, the claim goes to status *checked*.
- The details are stored in the claim table in the database.

openMIS 1.1.0 Insurees and Policies Claims Administration Tools Profile

Insuree enquiry...

< Claim XDDDFD33

Health Facility

JMHC001 Jamini Health Centre

CHF ID *

111111111

Name

Mister Family

Visit Date From *

2020-03-23

Visit Date To

Date Claimed *

2020-03-23

Visit Type *

Other

Main Diagnosis *

A009 Cholera, unspecified Cholera, uns

Claim No. *

XDDDFD33

Guarantee No.

Claimed

\$

Sec Dg1

Sec Dg2

Sec Dg3

Sec Dg4

Claim Administrator

JMHC0011 Lobo

Explanation

Policy Information

Insuree Last Visit

Code

Name

Expiry Date

Balance

Claim No.

Visit Date From

Visit Date To

Test sch

TST01

2020-01-31

\$ 0

test che

2020-03-20

Code

Name

Expiry Date

Balance

BCUL0001

Basic Cover Ultra

2021-01-31

\$ 0

Services

Service

Quantity

Price

Explanation

X

\$

Items

Item

Quantity

Price

Explanation

X

\$

REVIEW/FEEDBACK CLAIMS

- Contains only read-only data.
- Modifiable data of the feedback
 - Payment Asked
 - Drugs Prescribed
 - Drugs Received

openIMIS 10.0 Insurees and Policies ▾ Claims ▾ Administration ▾ Tools ▾ Profile ▾ Insuree enquiry.  

< Claim hjgfjgh  

Health Facility JMHC001 Jamini Health Centre	Insuree * Manth Gayle (111111116)	Date Claimed * 2019-12-04	Visit Date From * 2019-12-04	Visit Date To
Visit Type Other	Claim No. * hjgfjgh	Guarantee No.	Claim Status Checked	Feedback Status Idle
Review Status Idle				

Claim Administrator
JMHC0011 Lobo

Feedback

Feedback Date

Care Rendered



Payment Asked

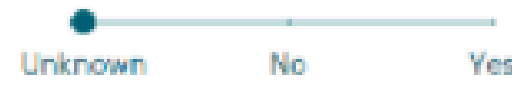


Claim Officer  

Drugs Prescribed



Drugs Received



Overall Assessment





REVIEW/FEEDBACK CLAIMS

- This functionality allows reviewing and adjustments of claims from the medical point of view.
- Reviewing of claims is restricted to users with the system role of Medical Officer.

openMIS 10.0 Insurees and Policies Claims Administration Tools Profile Insuree enquiry.

< Claim hjgfjgh

Health Facility	Insuree *	Date Claimed *	Visit Date From *	Visit Date To
JMHC001 Jamini Health Centre	Manth Gayle (111111116)	2019-12-04	2019-12-04	
Visit Type	Claim No. *	Guarantee No.	Claim Status	Feedback Status
Other	hjgfjgh		Checked	Idle
Review Status	Idle			
Claim Administrator	JMHC0011 Lobo			

Feedback

Feedback Date Claim Officer

Care Rendered

Unknown No Yes

Drugs Prescribed

Unknown No Yes

Payment Asked

Unknown No Yes

Drugs Received

Unknown No Yes

Overall Assessment

Unknown Very Bad Bad Not So Bad Not So Good Good Very Good

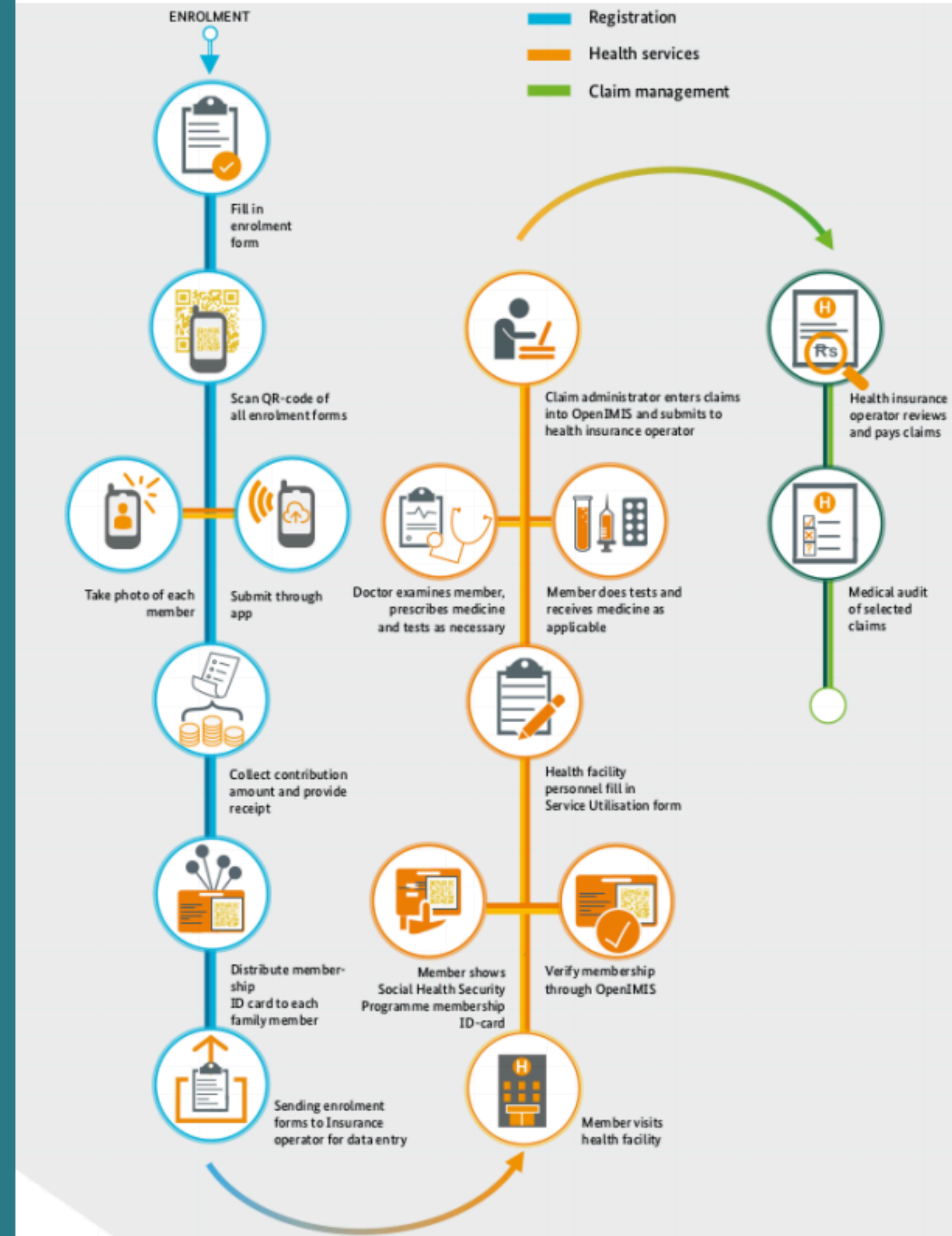


OpenIMIS in Nepal

OpenIMIS is being used to manage the major business processes of the Social Health Insurance (SHI) scheme in Nepal since 2016

Includes:

- door to door enrollment,
- electronic claims submission,
- review and adjudication,
- collecting feedback from beneficiaries.



REFERENCES

- <https://openimis.org/>
- <https://openimis.atlassian.net/wiki/spaces/OP/overview>
- <https://github.com/openimis>
- https://www.youtube.com/channel/UCujhZgz_6EFihAvYT_tD34Q



open**IMIS**

Thank You!